I B Tech, I Sem

Chemistry (R-23)

(CSE and CSM)

QUESTION BANK (10M) 2023-2024

S.NO	Questions	CO	BL	Marks
	MODULE-I STRUCTURE AND BONDING MODELS			
1	Derive the Schrodinger Time-independent wave equation.	1	3	10
2	Describe the basic features of Molecular Orbital theory.	1	2	10
3	Explain the molecular orbital diagram of Homo nuclear diatomic molecule.	1	2	10
4	Illustrate the MO energy level diagram of O_2 molecule with bond order and magnetic nature.	1	3	10
5	Derive the expression Schrodinger wave equation for Particle in one 1D Box?	1	2	10
6	Explain the π- Molecular orbital's of Benzene	1	2	10
7	Explain the molecular orbital diagram of Hetero nuclear diatomic molecule	1	2	10
8	a) Discuss de-Broglie hypothesis of matter waves. b) Describe Schrodinger Time dependent wave equation.	1	3	10
9	Explain the π- Molecular orbital's of 1, 3-Butadiene.	1	2	10
10	Interpret the bond order with the help of molecular energy level diagram of CO and predict the magnetic behavior.	1	2	10
	MODULE-II Modern Engineering Materials			
1	Explain Band theory of solids to classify the materials?	2	2	10
2	Distinguish the P-type and N-type of Semiconducting materials?	2	3	10
3	Describe P-N junction diode. Write any three uses?	2	2	10
4	Explain Semiconductor as a Rectifier?	2	2	10
5	Discuss P—N junction diode as a Transistor?	2	2	10
6	Define superconductors. Explain types, properties and Applications?	2	2	10
7	Describe super capacitors their classification, Working and Applications?	2	2	10
8	What are the Fullerenes and Explain their properties and Applications?	2	2	10
9	Write short note on carbon Nano tubes?	2	2	10
10	What are graphene? Explain their properties and Applications?	2	2	10

MODULE-III **Electrochemistry and Application**				
1	Derive the narnest equation?	3	2	10
2	Explain Galvanic cell? Write its application?	3	3	10
3	Write a short note on zinc-air battery? Applications?	3	2	10
4	Explain lithium-ion battery? Applications?	3	2	10
5	Write down H2-O2 fuel cell applications?	3	2	10
6	Explain conduct metric titration of HCL vs NAOH?	3	2	10
7	Explain conduct metric titration of weak acid with strong base (CH3COOH vs NaOH)	3	2	10
8	Write a short note on potentiometric titrations?	3	2	10
9	Calculate the reduction potential of Cu+2 in 0.01 at 250 c (E0 cu+2 /m= 0.337v)	3	2	10
10	Calculate the single electrode potentials of Zine in 0.001m Zinc sup;hat solution at 250(E0Zn/Zn+2=0.763v).	3	2	10
MODULE-IV **Polymers**				
1	Distinguish between Addition polymerization & condensation polymerization?	4	3	10
2	Explain cationic polymerization mechanism with suitable examples?	4	2	10
3	Explain free radical polymerization with examples?	4	2	10
4	Explain stereospecific /zeiglar-Natta (catalytic) polymerization with examples?	4	2	10
5	Differentiate b/w thermoplastics &thermosetting plastics?	4	3	10
6	Write prepacation, propostics, and applications of Bakelite?	4	2	10
7	Describe the following? a) PVC b) Polystyrene c) Teflon	4	2	10
8	Describe preparation, properties, applications of Buna-s, Buna-N rulers?	4	2	10
9	Define conducting polymers? Explain classification, application of conducting polymers?	4	2	10
10	Explain preparations, properties, applications of PLA, PGA?	4	2	10

UNIT-V Instrumental Analysis				
1	Explain the Lambert-Beer's law. Give its limitations	5	2	10
2	Describe the instrumentation and Applications of UV	5	4	10
3	Discuss the different types of electronic transitions	5	3	10
4	Define Electromagnetic spectrum? Explain the Basic Principle involved in U V Spectroscopy	5	2	10
5	Write a short notes on the following (a) Finger Print Region (b) Fermi resonance	5	3	10
6	Discuss the instrumentation and Applications of IR	5	3	10
7	Discuss the selection rules involved in I R Spectroscopy	5	3	10
8	Define Chromatography and Enumerate the classification of Chromatography with Examples	5	2	10
9	Illustrate the Principle and Instrumentation of HPLC	5	4	10
10	Write a short notes on the following (a) Applications of HPLC (b) Application Of I R Spectroscopy (c) Overtone band	5	3	10

Sign. of Course Co-ordinator **Sign. of HoD**